

Scaling from Metawebs to Realised Webs: A Hierarchical Approach to Network Ecology

Tanya Strydom ^{1*}, Alexander M. Dunhill ², Jennifer A. Dunne ³, Timothée Poisot ^{4,5}, Andrew P. Beckerman ¹

¹ School of Biosciences, University of Sheffield, Sheffield, UK, S10 2TN

² School of Earth, Environment and Sustainability, University of Leeds, Leeds, UK, LS2 9JT

³ Santa Fe Institute, 1399 Hyde Park Road, Santa Fe, NM 87501, USA

⁴ Université de Montréal, Département de Sciences Biologiques, 1375 Avenue Thérèse-Lavoie-Roux, Montréal, QC H2V 0B3, Canada

⁵ Québec Centre for Biodiversity Sciences, 1205 Docteur Penfield Avenue, Montreal, Quebec, Canada H3A 1B1, Canada

*Correspondence: t.strydom@sheffield.ac.uk

Abstract: Ecological networks provide a critical framework for understanding the architecture of biodiversity and predicting ecosystem responses to environmental change. However, the application of network ecology is often hindered by a lack of clarity regarding the assumptions inherent in different network representations. While recent literature predominantly focuses on data-centric classifications, predictive missing-link inference, or the topological specification of nodes and links, we focus instead on the theoretical gradients and processes that govern network construction. Here, we present a hierarchical framework that distinguishes between ‘metawebs’ (representing the fundamental feasibility of interactions) and ‘realised webs’ (representing interactions expressed in specific spatiotemporal contexts). We identify five core processes that drive the transition from potential to realised interactions: evolutionary compatibility and co-occurrence, which define the feasibility of links; and abundance, diet choice, and non-trophic interactions, which determine their realisation. Furthermore, we map these processes onto a methodological spectrum of network construction, distinguishing between inductive approaches that infer structure from observation, and deductive approaches that generate structure from mechanistic first principles. By making explicit the underlying biological assumptions and scale-dependent processes, this framework moves the field past descriptive topology, clarifying the scope of inference possible with each approach to facilitate more robust predictions of biodiversity dynamics in the anthropocene.

Keywords: food web, network construction, biodiversity, scale and process, interaction modelling, trophic network

1 Introduction

At the heart of modern biodiversity science are a set of concepts and theories about species richness, stability, and function, which have been discussed since the foundational work of [1], [2], and [3] and more recently [4]. These relate to the abundance, distribution, functions, and services that biodiversity provides. Network representations of interactions among organisms are increasingly argued to be an asset to understanding and predicting the impacts of multiple, simultaneous stress on biodiversity (*e.g.*, foundational studies on food web structure and robustness: [5–8]). Documenting interactions is thus one of the fundamental building blocks of community ecology and provides a powerful abstraction for mathematical and statistical modelling of biodiversity to make predictions, and to mitigate and manage threats [9].

However, there is a growing discourse around limitations to the interpretation and applied use of networks, which have been recognised since early discussions of sampling effort and aggregation in food webs ([5,10,11]; recent discussions: [12,13]). Against this, it is important to evaluate the value and the limitations of the various network conceptualisations and how these relate to biodiversity concepts, such as community structure or ecosystem function [14]. In this perspective we aim to provide an overview of different **food web** representations, particularly how each representation embeds assumptions about the processes that determine interactions (Section 3) about the levels of organization at which this occurs (*i.e.* the biological, ecological, spatial/temporal scales) and the way in which we construct the resulting networks (Section 4).

Network construction reflects both the data used and the theories governing species interactions. We still lack a clear explanation of the different assumptions and scale dependent processes that underpin network construction alongside extensive discussions about the challenges relating to data collection and observation [6,10,14–20]. Such an understanding should deliver an acceleration in capacity to more effectively predict the impact of multiple stressors on biodiverse communities.

In their recent work, [21] synthesised ecological networks based on node resolution and link type, with a focus on the data and methodologies underlying each representation. Here, we take a complementary theory-driven perspective by framing network construction as a hierarchical transition from feasibility (metawebs) to realisation. In the following sections, we review how nodes, edges, and interaction processes shape different network representations, highlighting how changes in assumptions alter ecological scale, node and

link resolution, modelling approaches, and the relative importance of evolutionary versus ecological processes. We conclude by aligning these representations with key questions in biodiversity science in the Anthropocene.

2 Setting the Scene: The Not So Basics of Nodes and Edges

Ecological networks serve multiple uses, representing an ‘object’ from which inferences can be made. While many aspects of community structure can be analysed without networks (*e.g.*, through trait distributions or abundance patterns) networks provide a formal framework for capturing the organisation of interactions among species. The study of network structure and topology has a long history in ecology, rooted in early theory on energy flow [2,22], and later extended to questions of robustness, stability, and complexity *e.g.*, [7,20,23–25]. More recent work has built on this foundation to link network structure to ecosystem functioning, persistence, and dynamical behaviour *e.g.*, [26–28]. Networks are therefore commonly treated as response variables in tests of ecological theory and statistical models of the generative processes that give rise to observed structure, and are widely used to compare communities across environmental gradients or through time [*e.g.*, 29,30]. They also provide a platform for evaluating downstream responses to perturbations, including secondary extinctions and robustness to species loss [*e.g.*, 7,31–33], as well as for implementing dynamical process that inference about stability, ecosystem function, invasions, climate change, contaminants, and extinction cascades [*e.g.*, 34,35,36]. Against this backdrop of multiple research agendas, the definition of ‘edges’ and ‘nodes’, and the levels of organisation at which they are defined, take many forms [17,37], each of which encode a series of assumptions within a network. Here we introduce a perspective on these baseline assumptions.

2.1 How do we define a node?

While nodes are conventionally described as representing species, in practice they may correspond to a range of taxonomic and non-taxonomic units, including sub-species, genera, or families, as well as trophic species (*e.g.*, [38,39]), feeding guilds (*e.g.*, [40]), or life-stage-specific subsets of species (*e.g.*, [41]). These choices reflect differences in the level, type, and consistency of resolution at which networks are constructed, rather than aggregation per se. In many cases, aggregation is applied explicitly on top of more highly resolved—or unevenly

resolved—data to address particular questions, as in the construction of trophic species, and the underlying data may remain available for alternative analyses. Nevertheless, representing nodes at coarser or mixed resolutions can limit taxon-specific inference (*e.g.*, whether species *a* consumes species *b*), bias estimates of degree distributions—particularly generality and vulnerability—and complicate downstream analyses such as extinction or invasion dynamics, where species identity and the consequences of loss may be obscured [41,42]. At the same time, there are clear justifications for using aggregated representations when the distribution of interactions among functional or trophic units is more informative than species-level detail, for example when analysing extinction patterns across feeding guilds [43]. More broadly, issues of resolution, scale, and sampling have long been recognised as central to the construction and interpretation of food webs (*e.g.*, [6,44]).

2.2 What is captured by an edge?

Understanding edges requires distinguishing between *potential* and *realised* links - potential links reflect feasibility, whereas realised links reflect fluxes such as energy transfer. Links within food webs are thus a representation of either potential links between species [45] or fluxes within a system *e.g.*, energy transfer or material flow as the result of the feeding links between species [2,46]. Edges can thus correspond to different ‘currencies’ [21]. There are also a myriad of ways in which the links themselves can be specified. Links between species can be treated as present or absent (*i.e.*, binary), may be defined as probabilities [47,48] or by continuous functions which further quantify the strength of an interaction [49]. Link definition depends on both the ecological currency and how interactions are represented. For example, feasibility is unlikely to accommodate flux, but does align with binary or probability representations. Taking a food web that consists of links representing feasible interactions among a collection of species will be meaningless if one is interested in understanding the flow of energy through the network as the links are not environmentally/energetically constrained.

2.3 Network representations

Given these definitions of nodes and edges, ecological networks can be divided into two broad types: **metawebs**, which represent all *potential* interactions within a species pool [44], and **realised networks**, which represent the subset of interactions expressed within a particular community at a given place and time. These representations differ in both

the scales at which they are constructed and the processes assumed to generate their structure.

A metaweb is fundamentally a list of *feasible* interactions between species. Feasibility is determined by trait complementarity, typically related to feeding, and can be further constrained by species co-occurrence, producing a transition from *global* to *regional* metawebs. Metawebs therefore identify evolutionarily and regionally plausible interactions, ecologically impossible (*i.e.*, forbidden) links [50], and the potential diet breadth of species [51].

In contrast, realised networks are localised in space and time, with links shaped by co-occurrence, environmental conditions, and diet choice. Even when represented as binary matrices, their links implicitly reflect interaction strength and energy transfer. Realised networks are therefore not simple spatial or temporal subsets of metawebs; they emerge from the processes governing whether feasible interactions are expressed. Consequently, a metaweb and realised network containing the same species may differ substantially in structure because link presence is governed by different constraints. Links absent from a metaweb represent infeasible interactions, whereas links absent from a realised network reflect context-dependent ecological constraints.

3 From Nodes and Edges to Process and Constraints

In the previous section we discussed how the definition of nodes and edges, representing different scales and processes, lead to the concept of a metaweb and a realised web. The fundamental take-homes are that nodes vary in their resolution, edges vary in what kind of process they represent and the intersection of these, defined by meta- vs. realised webs, underpins distinct lines of inquiry and constraints on the type of inference we can make with networks. Here we reveal five core constraints across evolutionary and ecological scales that further delineate the transition from meta- to realised webs, exposing processes that determine the nature of links among nodes: evolutionary compatibility, co-occurrence, abundance, diet choice, and non-trophic interactions Figure 1.

[Figure 1 about here.]

3.1 Processes that determine the feasibility of an interaction

Evolutionary compatibility and co-occurrence are the two principal processes that ‘act’ at the species pair of interest and define feasibility. The scale of inference and set of processes embodied in these two constraints typically combine to define a ‘list’ of interactions that are viable/feasible and defined strictly as present/absent. Reflecting on the previous section, nodes are typically species and rules defining edges are defined by trait complementarity (phylogenetic) and/or co-occurrence. Here we provide more insight into each process.

Evolutionary compatibility

This constraint is defined by shared (co)evolutionary history between consumers and resources [52–55] which is manifested as ‘trait complementarity’ between two species [56]. In this body of theory, the consumer has the ‘correct’ set of traits that allow it to acquire and consume the resource. Interactions that are not compatible are defined as forbidden links [50]; *i.e.*, they are not physically possible and will *always* be absent within a network. Networks do not properly arise from models based on this constraint. Instead, interacting species pairs are defined and these are represented as binary (possible vs forbidden) or probabilistic [47]. For example, in the metaweb constructed by [57] probabilities are quantified as the confidence of a specific interaction being *possible* between two species. A network constructed on the basis of evolutionary compatibility is conceptually aligned with a ‘global metaweb’, and gives us information as to the global feasibility of links between species pairs despite the fact that they do not co-occur (see Figure 1).

(Co)occurrence

The co-occurrence of species in both time and space is a fundamental requirement for an interaction between two species to occur (at least in terms of feeding links). Although co-occurrence data alone is insufficient for building an accurate and ecologically meaningful representation of *feeding links* [58], it is still a critical process that determines the possible realisation of a feeding. Knowledge on the co-occurrence of species allows us to spatially constrain a global metaweb to reflect regional metawebs [59]. In the context of Figure 1 this would be the metawebs for regions one and two.

We reinforce that these two constraints don’t deliver a network *per se*, but a list of feasible species pairs. Although it is possible to build a network from the list of interactions generated by these constraints, it is important to be aware that the structure of this network is not constrained by any community context - just because species are able to

157 interact does not mean that they will [60,61].

158 3.2 Processes that realise networks

159 In contrast to the above, here we highlight three processes that influence the *realisation* of
160 an interaction between species and thus form the conceptual basis for realised networks.
161 As we show in Figure 1, a ‘truly realised’ network is the product of properties of the
162 community (**abundance** and **non-trophic interactions**) and the individual (**diet**
163 **choice**). This represents a conceptual shift from considering the feasibility for species
164 pairwise interactions to considering the edge as a representation of energy flow. Such
165 a transition requires information about how the community, the environment and the
166 individual *constrains* network topology as defined by consumer choice ([62], Section 2.3)

167 Abundance

168 Abundance as a realising process emerges from a null model for energy acquisition:
169 organisms feeding randomly will consume resources in proportion to their abundance
170 [63]. Here, abundance of different prey species influences the distribution of links in a
171 network [64] by defining a preference linked to individuals among species meeting [47,60].
172 Abundance data (linked to a derived metaweb) delivers a foundation ruleset that can
173 define the distribution and strength of links. Of note, however, is that such abundance
174 constrained interactions are not necessarily contingent on there being any compatibility
175 between species [65–67].

176 Diet choice

177 It is well established that consumers make more active decisions than eating items in
178 proportion to their abundance [63]. Ultimately, consumer choice is underpinned by
179 an energetic cost-benefit framework centered around profitability and defined by traits
180 associated with acquisition and consumption of a resource [68,69]. Energetic constraints
181 are invoked to construct networks in a myriad of ways [*e.g.*, 42,70–72].

182 Unlike metaweb approaches, these models generate realised webs as emergent outcomes of
183 consumer behaviour. We also here make a distinction, developed below, with models like
184 the Niche Model [39], where diet choice is implicit in its probabilistic network generating
185 function, but it is working to replicate the *expected* structure of the network, and this
186 structure does not emerge from node-based rules. Note that we select diet choice as a
187 term to capture rules linked to optimal foraging [73] and metabolic theory [74]; it is a

188 sensible ‘umbrella concept’ for capturing the energetic constraint on of the distribution
189 and strength of interactions.

190 **Non-trophic interactions**

191 We include non-trophic interactions [see 75] here not as a determinant of links, but a
192 modifier of them - they are the community context above and beyond co-occurrence and
193 abundance. Non-trophic interactions include competition for space, predator interference,
194 refuge provisioning, recruitment facilitation as well as non-trophic effects that increase or
195 decrease mortality. These interactions specifically modify either the realisation or strength
196 of trophic interactions [28,31,76–78] and represent direct (*e.g.*, predator *a* outcompetes
197 predator *b*) and indirect (*e.g.*, mutualistic/facilitative interactions) mechanisms.

198 Some interactions, such as pollination, occupy an intermediate position in this framework,
199 as they combine trophic components (*e.g.*, resource consumption) with non-trophic effects
200 that influence reproduction, recruitment, and population persistence [79–81]. They operate
201 on the realisation of a network by altering the fine-scale distribution and abundance of
202 species and relative contributions of direct and indirect effects to biomass, persistence,
203 stability and the functioning of the communities [75,82–84].

204 **4 Network construction**

205 The above five processes are central to understanding the assumptions inherent in building
206 different types of networks. Each of the processes, or combinations thereof, deliver a
207 unique set of boundary conditions on what a network represents and can be used for. Here
208 we build on the introduction of these five processes to further categorise the approaches to
209 constructing networks. In doing so also introduce more detail on a variety of methodologies
210 used to construct networks.

211 **4.1 Why construct networks?**

212 Networks are a representation of biodiversity. In a perfect world, we might know about
213 all interactions. However, the empirical collection of interaction data is both costly and
214 challenging to execute [50,85,86]. In the absence of robust empirical data, we construct
215 models that facilitate interpolation and gap-filling of existing empirical datasets [*e.g.*,
216 87,88–90], predict the feasibility of interaction among pairs of species, or directly predict
217 network structure [see 91 for a broader discussion].

218 They are unique in delivering more than just estimates of species richness. As noted in
 219 the introduction, a network embodies the organising structure of biodiversity and allows
 220 numerous opportunities for ‘downstream’ analysis, including the comparison of structures,
 221 estimation of energy flux or extinction dynamics and ultimately form the structural
 222 inputs to dynamical systems models that facilitate ecological and conservation relevant
 223 inference about productivity-diversity-stability-function relationships [27] in space and
 224 time. But making such inferences requires careful attention to one or more of the processes
 225 discussed in Section 3. While these network representations may be simplifications of
 226 the truth [92], they remain critically useful tools as they allow us to move beyond simple
 227 descriptions of species richness to test hypotheses about community architecture and
 228 ecosystem functioning that would be otherwise impossible to assess.

229 **4.2 Construction through induction**

230 Constructing feasible or realised networks can be framed as an ‘inductive reasoning’
 231 process where insight and generalisation arises from a set of observations and relationships.
 232 Inductive reasoning as a foundation for network construction is implemented at node
 233 and network levels. When applied at the node level, species-specific networks are created
 234 and judged by their association with expected feeding interactions. When applied at the
 235 network level, networks are judged by their structural properties.

236 **4.2.1 Species specific networks: construction through node level induction**

237 Constructing feasible networks and facilitating the interpolation or gap-filling of existing
 238 empirical datasets on sets of species interactions can be framed as an ‘inductive reasoning’
 239 process where insight and generalisation arises from a set of observations and relationships
 240 about feeding. All methods in this inference space rest on a set of three assumptions:
 241 there are a set of ‘feeding rules’ that underpin interaction feasibility [93]; these rules
 242 are phylogenetically conserved [54,94]; and they can be specified by matching the traits
 243 between consumer and resource.

244 Evolutionary compatibility and co-occurrence constraints have been critical to the construc-
 245 tion of ‘first draft’ networks for communities for which we have no interaction data [57].
 246 They are also central to interpolation in data poor regions and predicting interactions for
 247 ‘unobservable’ communities *e.g.*, prehistoric networks [43,95,96] or future, novel community
 248 assemblages [97]. Furthermore, they have the capacity to evaluate a role of interactions
 249 among species relative to their distribution by accounting for the role of the environment

and the role of species interactions [98–101]. There are substantial data requirements for these approaches including expert knowledge, species traits and phylogenetic relationships and/or interaction data on related species or communities.

Feeding rules can be specified in several ways. Expert-based approaches define feasible interactions using trait matching [102,103], while mechanistic approaches often rely on body-size relationships between consumers and resources [104,105]. Alternatively, statistical and machine-learning methods infer interaction probabilities from observed networks using ecological predictors such as traits and phylogeny [106].

Rules are also defined by correlating real world interaction data with suitable ecological proxies for which data is more widely available (*e.g.*, traits) using some sort of binary classifier (see [106] for an overview). These include generalised linear models [*e.g.*, 107], random forest [*e.g.*, 108], trait-based k-NN [*e.g.*, 109], and Bayesian models [*e.g.*, 110,111].

Finally, graph embedding uses the structural features of a known network to infer the position of species in an unknown network through the decomposition of the interaction onto the embedding space (see [51]). This decomposition relies on a combination of ecological proxies (*e.g.*, phylogenetic relatedness [57]) in conjunction with known interactions to infer the latent values of species, which can then be mapped onto decomposition of a known network.

4.2.2 Species agnostic networks: construction through structure induction

These models generate ecologically realistic network structures using simple probabilistic rules. They are commonly used as null expectations, for comparative analyses, and as inputs to dynamical models. The determination of links between species is not directly linked to properties of the nodes. This means these networks are usually not species specific. Although they require little empirical information, they encode explicit assumptions about expected network structure.

Stochastic network models use a probabilistic rule-set about diet choice and niche breadth to reflect fundamental ideas of foraging biology. These models that are based on the compartmentalisation and acquisition of energy for species at different trophic levels [112,113] and that network structure can be determined by distributing interactions along single dimension [the ‘niche axis’; [114]]. Typically these models parametrise some aspect of the network structure [although see 112 for a parameter-free model]. These models include the most commonly used network generator, the Niche model [39], as well as the

original Cascade model [5] and the derived Nested hierarchy model [115]. Despite the fact that these networks are derived without any real world data they are still able to recover the structure of emirical networks [116]. These models often form the basis for dynamic models *e.g.*, the allometric trophic network [24,26] and bioenergetic food web models [34].

4.3 Construction through deduction

In contrast to the above approaches centred on feasibility, realised networks via methods reflecting abundance and diet choice typically rely on deductive reasoning and have a unique agenda to those above. In contrast to the inductive methods, inference about a realised network follows from a set of premises defining generative processes, often referred to as mechanisms, allowing inference about the distribution and strength of interactions. These models are widely used to study environmental gradients, energy flux, extinction dynamics, and ecosystem functioning. They also provide the structural backbone for dynamical systems modelling to address questions about stability-structure-productivity-function relationships, secondary extinction dynamics, species invasion and climate change. There are two broad groups of models in this deductive category.

4.3.1 Species-specific networks

These models capture the behaviour of the nodes by explicitly taking into account the properties of the different species in the community. Which means that there is a degree of variance in which links are predicted between species unlike the more ‘static’ predictions made by inductive models. However, these networks are ‘costly’ to construct in real world settings (requiring data about the entire community, as it is the behaviour of the system that determines the behaviour of the part) and also lack the larger diet niche context afforded by metawebs.

Neutral networks are built on the assumption that foraging decisions are tied *only* to the abundance of species within the community [117,118]. Here links are solely determined by the relative abundance of the different species in the community. Although unrealistic as a complete explanation, neutral models can be combined with inductive approaches to generate more localised predictions [67].

There is a broader group of models that focus on determining interactions in terms of energetic constraints on diet breadth, often using the ratio of consumer-resource bodysize as a proxy for capturing the energetic constraints of feeding. Models such as those developed

by [71] and [68] are similar to the mechanistic approaches discussed in Section 4.2, however instead of determining interactions based on mechanistic feasibility it is rather constrained by the energetic cost of predation. Note that although these models do not place any explicit constraints on the expected structure of the network, the links should still be considered as ‘realised’ owing to the energetic constraint placed on links. A different subset of diet models [*e.g.*, 42,119] use a diet choice approach, however similar to the stochastic network models they also embed assumptions on network structure. Thus these models predict both interactions and network structure simultaneously, although they would benefit in being refined by more explicitly accounting for trait-based (*i.e.*, feasibility) parameterisation [35].

5 Making Progress with Networks

The motivation to leverage network ecology in conservation ecology, environmental risk assessment and natural resource management stems from a shift away from species/population specific measures of the effects of stress and disturbance to community level metrics of these impacts. These metrics, such as resilience and more generally stability, ecosystem function and biodiversity *per se*, are natural properties of networks. This suggests that modern conservation, risk assessment and resource management requires robust network tools to support decision making.

This is also true in the disciplines of ecology and environmental science and their focus on abundance, distribution, functions and services that biodiversity provides [4]. Major questions remain, for example, about stability-diversity-productivity relationships, the impacts of extinctions and invasions and the impacts of multiple stressors operating at multiple ecological scales. A network approach to answering these types of questions specifically allows us to evaluate how environmental gradients and anthropogenic stress map through direct and indirect effects among species in a complex community and reveal fundamental patterns and understanding of processes in the natural world.

In order to effectively use networks to aid us in answering questions about conservation/risk assessment/management and core ecological theory, we need to be mindful that we are mapping the *correct* network representation to the question of interest [21]. Notably, there are certain questions that cannot be answered using specific network representations as the scale of the question of interest is fundamentally misaligned with either the process captured by a specific network representation Section 3.1, the underlying data that is used

345 to construct it Section 4 or both of these factors.

346 Here we discuss and map the different network representations shown in Figure 1 to
 347 ‘appropriate’ research questions and agendas see also 1. We also highlight some of the key
 348 methodological challenges that currently limit our conceptualisation of a ‘network’ and
 349 thus impact their effective practical application in real world settings.

Table 1: Showcasing some of the broader avenues of inquiry, specifically how they map to the different network representations. Additionally we highlight some studies that address or present opening discussions around each research question. Superscripts at the research question indicate the strength of the current literature in addressing that research question: $\checkmark$ indicates area with strong foundational research, Δ partial/emerging areas of research, and $\times$ areas where research is weak/largely absent

Network		
Representation	Example Research Question	Representative Studies
Global Metaweb	How will novel communities respond to <i>e.g.</i> , extinction, turnover, invasion and rewilding $\checkmark$	[120]; [7]
	Diet-based conservation focusing not only on the target species but the species it might depend on for food resources Δ	[121]; [122]; [123]
	Rewiring capacity/potential of species by looking at the <i>entire</i> diets of species Δ	[124]; [31]; [125]; [126]
	Eco-Evolutionary dynamics and how they relate to the conservation and origination of feeding strategies $\times$	[60]; [127]
Regional Metawebs	Applied use potential of questions highlighted for global metawebs at the management scale <i>e.g.</i> , a protected area Δ	[101]; [128]; [129]

Network		
Representation	Example Research Question	Representative Studies
Realised webs	Refinement/extension of species distribution models by incorporating co-occurrence and species associations <i>e.g.</i> , predator and prey ✓	[130]; [131]
	The allocation of multiple stressors across networks Δ	[132]; [133]
	Temperature threshold to community collapse Δ	[134]; [135]
	Extinction and persistence after harvesting/invasion/extinction ✓	[136]; [137]
	Stability-diversity-productivity-function ✓	[138]; [121]
	Explicitly tying ecosystem level processes and nutrient flows to networks $\times$	[139]
	Meta communities and the idea of meta-network-communities Δ	[140]; [141]

5.1 Key Eco-Evo-Conservation Questions

5.1.1 Global Metawebs

Global metawebs are most appropriate for questions centred on interaction feasibility and potential diet breadth. They provide a platform for exploring hypothetical or novel communities under climate change [142], species invasions, reintroductions, and rewilding, and the potential rewiring capacity [126]. Because they focus on feasible rather than realised interactions, global metawebs are also well suited to studying eco-evolutionary dynamics and how evolutionary history, natural selection, and phenotypic plasticity shape interaction niches.

359 5.1.2 Regional Metawebs

360 Regional metawebs extend these questions by explicitly incorporating species co-occurrence
361 and therefore provide a more management-relevant perspective. They can be used to
362 refine species distribution models and projections of future community composition [29,40].
363 However, caution is required when comparing regional metaweb structure across space
364 or environmental gradients, as observed differences may reflect species turnover (*e.g.*,
365 β -diversity) rather than changes in interaction processes.

366 5.1.3 Realised networks

367 Realised networks are best suited to questions concerning how community and environmen-
368 tal context shape interaction structure and ecosystem functioning [143]. They provide the
369 appropriate framework for studying stability, resilience, biodiversity dynamics, ecosystem
370 functioning, extinction cascades, invasions, climate change impacts, and network rewiring
371 through time. By capturing interactions that are actually expressed, realised networks
372 allow investigation of how perturbations propagate through communities and influence
373 persistence.

374 The increasing availability of long-term interaction datasets is expanding opportunities
375 to address these questions [27,144]. However, empirical datasets often accumulate in-
376 teractions across extended periods, potentially obscuring temporal variation in realised
377 interactions [10]. Developing approaches that better reconcile empirical networks with
378 realised community dynamics therefore remains an important challenge.

379 5.2 Key methodological challenges

380 As noted above, the three network types highlight longstanding methodological challenges
381 that limit both the precision and accuracy of ecological inference. Here we briefly review
382 these challenges and emerging opportunities to address them.

383 **Understanding what empirical data represents:** Robust inference requires under-
384 standing what constitutes an observed interaction, whether recorded directly (predation
385 events) or indirectly (*e.g.*, gut contents or stable isotopes). Because empirical networks
386 often accumulate observations across space and time, they may be conceptually closer to
387 metawebs than realised networks.

388 **The validation of network structure:** Considerable progress has been made in
389 assessing how well models recover pairwise interactions [91,145], yet there remains no clear

framework for evaluating their ability to recover network structure [114,146]. This raises two related questions: what constitutes an appropriate benchmark, and which aspects of network recovery matter most? For metawebs, accurately identifying both present and forbidden links may be essential, whereas for realised webs it remains unclear whether recovering pairwise interactions, aggregate properties (*e.g.*, connectance), or both should be the primary objective.

Transitioning between metawebs and realised webs: Most approaches for modelling realised networks do not explicitly incorporate evolutionary constraints (although see [147]; [68]). Progress will likely require either ensemble approaches [148,149] or methods for downscaling metawebs into realised networks [e.g., 150]. However, any such transition must retain clarity about the meaning of links - structurally realistic networks do not necessarily represent realised prey choice or energy flow.

Developing frameworks that allow transitioning between metaweb and realised representations will also facilitate integration with metacommunity, metaecosystem, and ecosystem-level theory [151,152]. Doing so requires clear definitions of what constitutes a network and where its spatial and temporal boundaries lie [153].

Making networks more tractable in applied spaces: The application of networks to conservation and management remains limited by difficulties in defining network boundaries, aligning them with management units, and interpreting network metrics in policy-relevant ways [154]. Addressing these challenges will require stronger links between network structure and ecosystem function, alongside careful matching of analytical tools and network representations to management objectives [128,155].

Taken together, these challenges highlight three overarching messages. (i) Network representations are inseparable from the data and assumptions used to construct them; (ii) validation and benchmarking must be aligned with the intended network type and research question; and (iii) greater conceptual clarity is needed when transitioning between metawebs and realised networks, particularly in applied contexts. Explicitly articulating these distinctions is essential if networks are to be used rigorously and transparently across scales.

6 Conclusion

1. **Network representations encode different processes:** Different network representations capture distinct assumptions, data structures, and ecological processes; understanding the interplay between a network’s structure and the processes it encodes is critical for determining which questions it can address.
2. **Network representations are not interchangeable:** There is no universally optimal network representation. The utility of a network depends entirely on how well its underlying assumptions and data align with the intended scale and process of analysis.
3. **Explicit assumptions are essential for evaluation:** Explicitly articulating the assumptions behind network assembly facilitates a more critical evaluation of network suitability and inferential power. Consequently, researchers can move beyond the uncritical use and default reliance on conventional network representations.
4. **A unified framework improves application and progress:** The framework presented here provides a structured basis for comparing network representations and evaluating their suitability across conceptual, methodological, and applied contexts. Establishing this standard is essential for preventing the misinterpretation of network data and for addressing foundational questions in biodiversity science.

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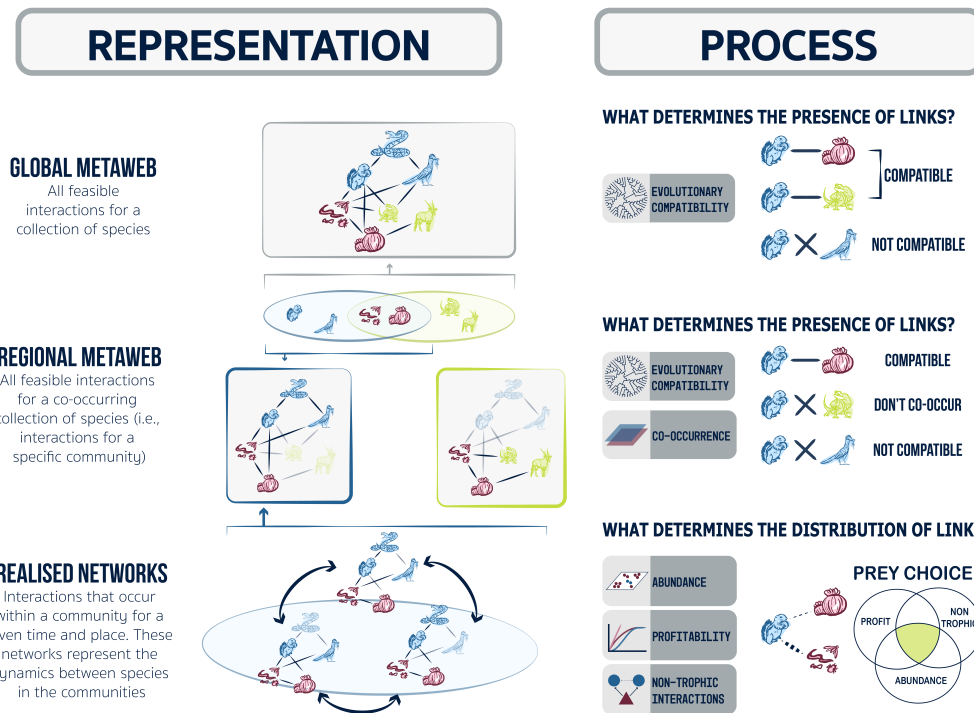


Figure 1: Aligning the processes that determine interactions (right) with different network representations (left). A **global metaweb** captures all feasible interactions among a species pool. Restricting this network by species co-occurrence yields **regional metawebs**, which differ in species composition and interaction feasibility. Species occurring in both regions are shown in red, whereas region-specific species are shown in blue and yellow. **Realised networks** represent subsets of regional metawebs expressed in particular spatial and temporal contexts. Their structure is shaped not only by co-occurrence, but also by community-level processes such as abundance, diet choice, and non-trophic interactions.